COMPUTER ENGINEERING

QUEZON CITY

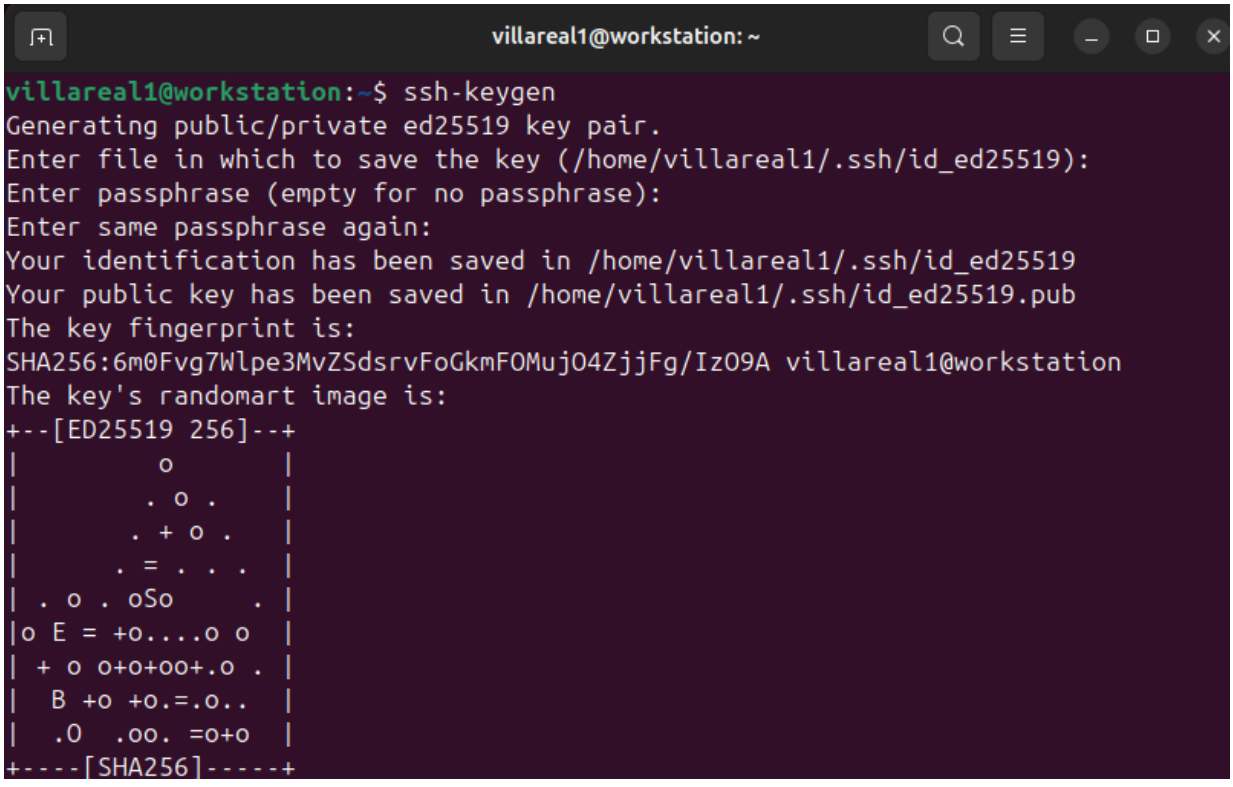
COURSE CODE/TITLE:	CPE-212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	07/16/26	**DATE SUBMITTED:**	07/30/26
NAME:	Villareal, Vanderohe Vernon	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 2.1 **SSH Key-Based Authentication and GIT Setup**

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication

1. The simplest way to generate a key pair is to run ssh-keygen without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user authentication are usually stored in the users .ssh directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case id_rsa when using the default RSA algorithm. It could also be, for example, id_dsa or id_ecdsa.



2. Issue the command ssh-keygen -t rsa -b 4096. The algorithm is selected using the -t option and key size using the -b option.

```
villareal1@workstation:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/villareal1/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/villareal1/.ssh/id_rsa
Your public key has been saved in /home/villareal1/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:2AejzWvbAzZ8fWB5sgalctM9H8ILDS4nlYjmbzHybPc villareal1@workstation
The key's randomart image is:
+---[RSA 4096]----+
|        . . .    |
|       o . +.    |
|      o o o++o   |
|       O.B=+*++. |
|      o.SoB=.=oo.|
|        =O..+.. .|
|       .=+....   |
|       . o.  E   |
|        . ..     |
+----[SHA256]-----+
```

3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that it cannot be used even if someone obtains the private key file. The passphrase should be cryptographically strong.

```
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
```

4. Verify that you have created the key by issuing the command ls -la .ssh. The command should show the .ssh directory containing a pair of keys. For example, id_rsa.pub and id_rsa.

```
villareal1@workstation:~$ ls -la .ssh
total 32
drwx------  2 villareal1 villareal1 4096 Jul 16 02:07 .
drwxr-x--- 15 villareal1 villareal1 4096 Jul 15 23:48 ..
-rw-------  1 villareal1 villareal1    0 Jul 15 23:41 authorized_keys
-rw-------  1 villareal1 villareal1  419 Jul 16 02:04 id_ed25519
-rw-r--r--  1 villareal1 villareal1  104 Jul 16 02:04 id_ed25519.pub
-rw-------  1 villareal1 villareal1 3389 Jul 16 02:07 id_rsa
-rw-r--r--  1 villareal1 villareal1  748 Jul 16 02:07 id_rsa.pub
-rw-------  1 villareal1 villareal1 1546 Jul 16 01:41 known_hosts
-rw-r--r--  1 villareal1 villareal1  142 Jul 16 01:26 known_hosts.old
```

Task 2: Copying the Public Key to the remote servers

1. To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh- copy-id tool.

2. Issue the command similar to this: ssh-copy-id -i ~/.ssh/id_rsa user@host

```
villareal1@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa villareal1@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/villareal1/
.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
villareal1@server1's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'villareal1@server1'"
and check to make sure that only the key(s) you wanted were added.

villareal1@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa villareal1@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/villareal1/
.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
villareal1@server2's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'villareal1@server2'"
and check to make sure that only the key(s) you wanted were added.
```

3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.

4. On the workstation, verify that you can SSH with server1 and server2. What did you notice? Did the connection ask for a password? If not, why?

```
villareal1@workstation:~$ ssh villareal1@server1
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Thu Jul 30 00:02:00 2026 from 192.168.56.134
```

```
villareal1@workstation:~$ ssh villareal1@server2
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Thu Jul 30 00:10:48 2026 from 192.168.56.133
```

- The workstation no longer asks for a password in order to run an ssh command of a remote server, since now that the server has a public key it would use and encrypt that key into the private key of the workstation and it of matches it would prove that its an identity which grants you access instantly.

Task 3: Set up the Git Repository

1. On the local machine, verify the version of your git using the command which git. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: sudo apt install git

```
villareal1@workstation:~$ sudo apt install git
[sudo] password for villareal1:
Sorry, try again.
[sudo] password for villareal1:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer require
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitw
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 4,806 kB of archives.
After this operation, 24.5 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl a
7029-2 [25.6 kB]
```

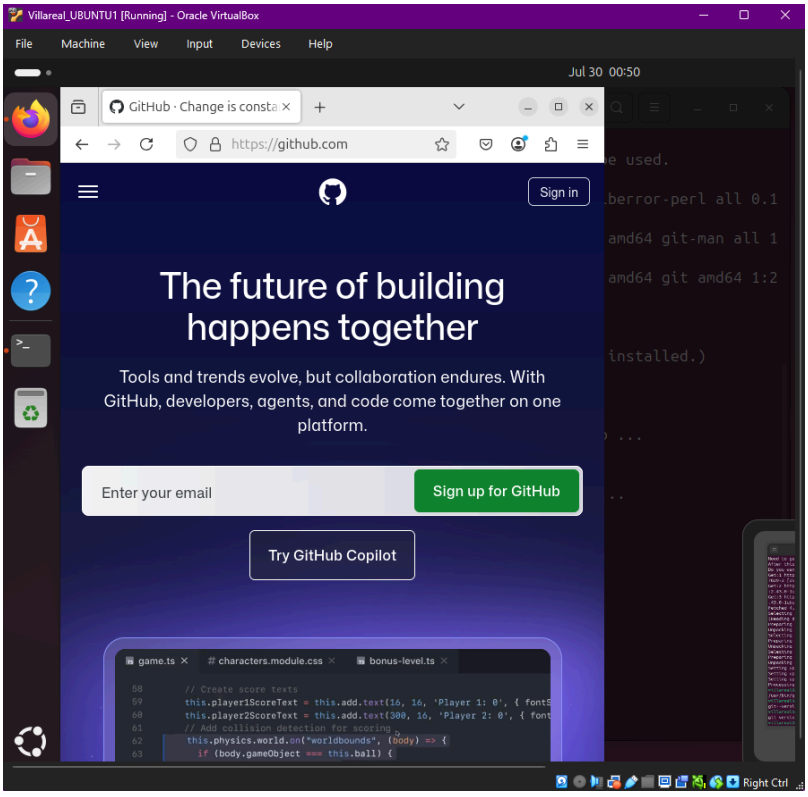
2. After the installation, issue the command which git again. The directory of git is usually installed in this location: user/bin/git.

```
villareal1@workstation:~$ which git
/usr/bin/git
```

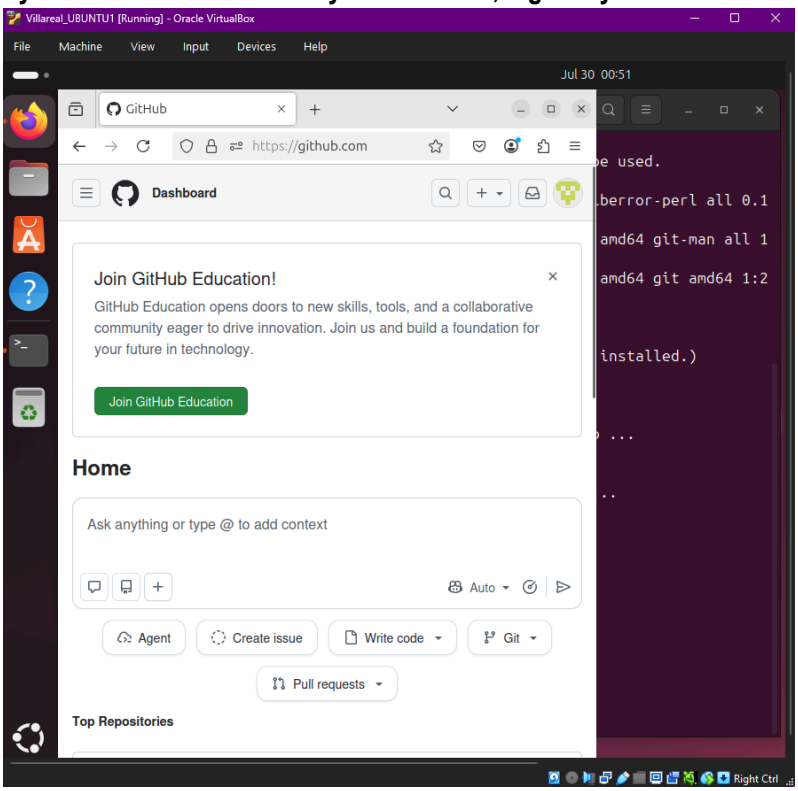
3. The version of git installed in your device is the latest. Try issuing the command git --version to know the version installed.

```
villareal1@workstation:~$ git --version
git version 2.43.0
```

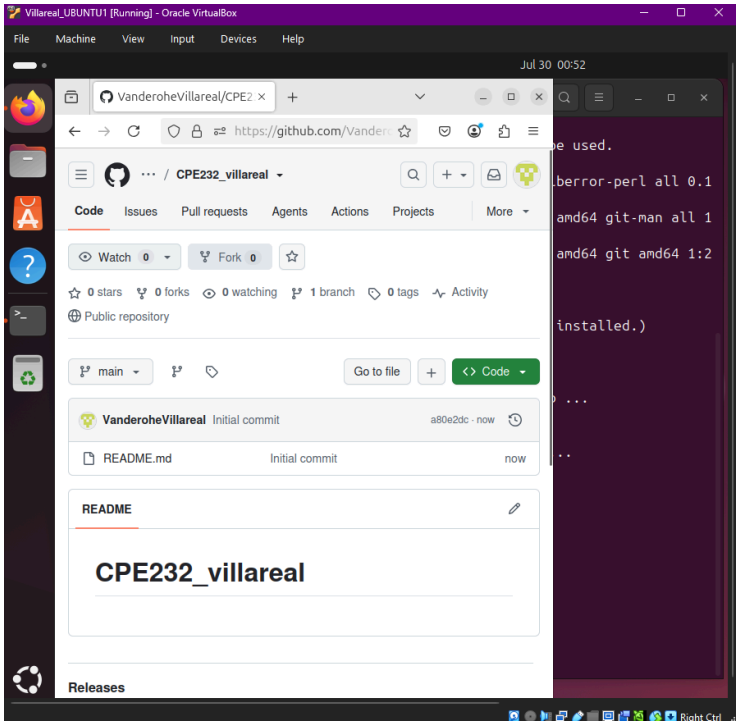
4. Using the browser in the local machine, go to www.github.com.



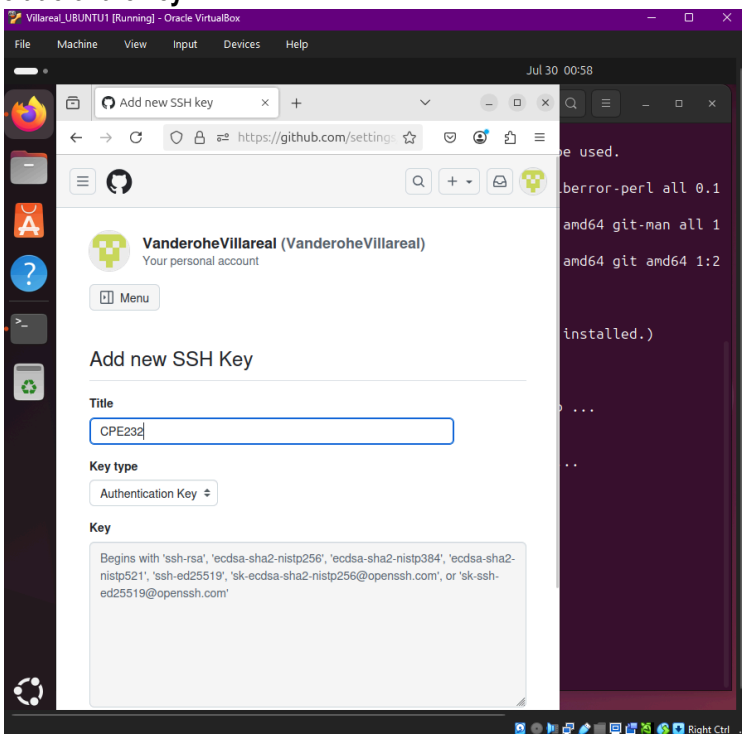
5. **Sign up in case you don't have an account yet. Otherwise, login to your GitHub account.**



a. Create a new repository and name it as CPE232_yourname. Check Add a README file and click Create repository.



b. Create a new SSH key on GitHub. Go to your profile's settings and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.

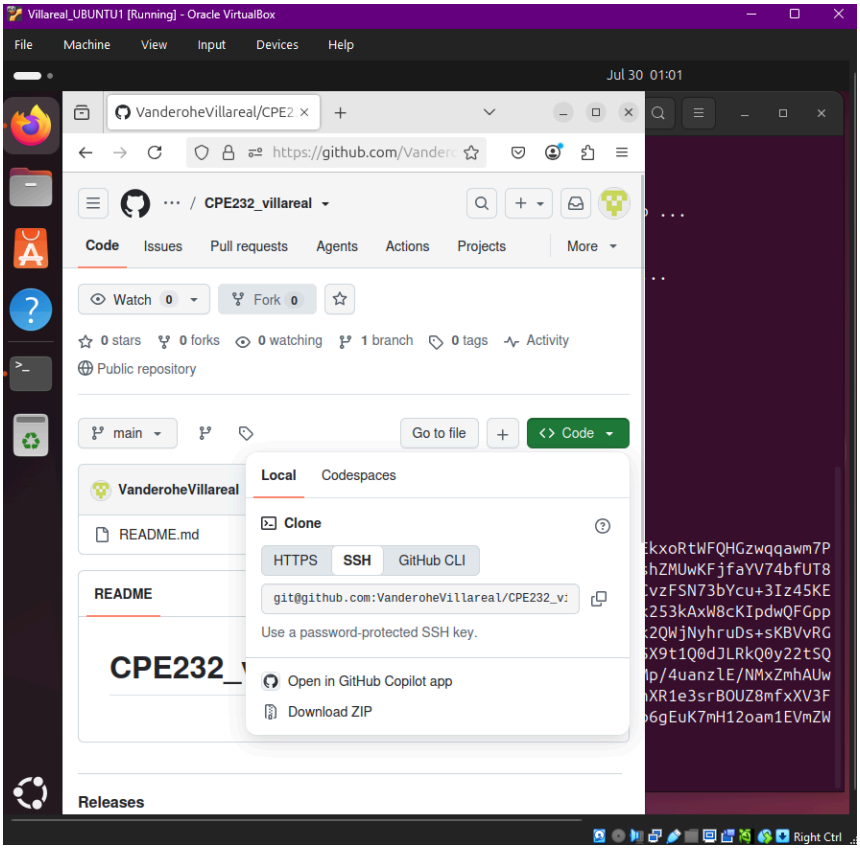


c. On the local machine's terminal, issue the command cat .ssh/id_rsa.pub and copy the public key. Paste it on the GitHub key and press Add SSH key.

Key

```
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAACAQCP8NcrCH77BqC8r7m5ue1kEkxoRt
WFQHGzwqqawm7PAQSdmPY/6hJX1KXKV+lYqiiSmipupJhQ6xUB51FAdOFqh
UySHXZ+Xs9zE3jlshZMUwKFjfaYV74bfUT84XVajElqw1cTFp1OkK1XwAtLtSU2
ypKS9SJQva0GT3EGbigNEsZfcajc0nk/
CvzFSN73bYcu+3Iz45KEwJWVMYtrozlUCJDxnafW8XLF1AXbEKtwzh9ZNxd6at
+W7p36byUorwX8S87mk253kAxW8cKIpdwQFGppWWNGK9rWwiM4zabTBVtU
pRiNnELjZ/Sv/
WtgHoHtxBjPoe4lmKEY0HLmfaD+k2QWjNyhruDs+sKBVvRG3ZWCO1Zq3C02
```

d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the Code drop down menu. Select SSH and copy the link.



e. Issue the command git clone followed by the copied link. For example, git clone git@github.com:jvtaylar-cpe/CPE232_yourname.git. When prompted to continue connecting, type yes and press enter.

```
villareal1@workstation:~$ git clone git@github.com:VanderoheVillareal/CPE2
lareal.git
Cloning into 'CPE232_villareal'...
The authenticity of host 'github.com (20.205.243.166)' can't be establishe
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvC
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hos
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

f. To verify that you have cloned the GitHub repository, issue the command ls. Observe that you have the CPE232_yourname in the list of your directories. Use CD command to go to that directory and LS command to see the file README.md.

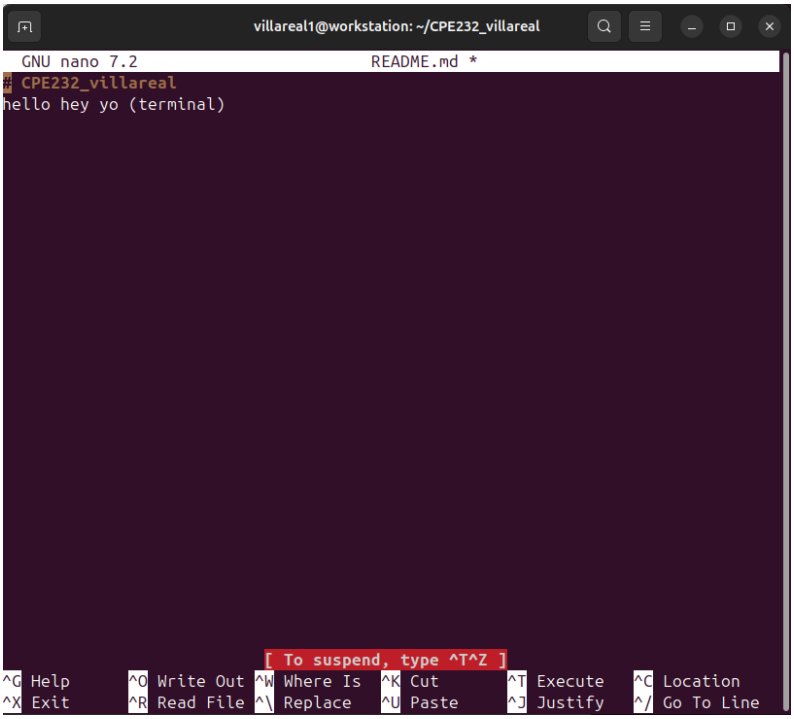
```
villareal1@workstation:~$ ls
CPE232_villareal  Documents  Music     Public  Templates
Desktop           Downloads  Pictures  snap    Videos
```

g. Use the following commands to personalize your git.

- git config --global user.name "Your Name"
- git config --global user.email yourname@email.com
- Verify that you have personalized the config file using the command cat ~/.gitconfig

```
villareal1@workstation:~$ git config --global user.name Villareal
villareal1@workstation:~$ git config --global user.email qvvvillareal@tip.edu.ph
villareal1@workstation:~$ cat ~/.gitconfig
[user]
        name = Villareal
        email = qvvvillareal@tip.edu.ph
```

h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.



i. Use the git status command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven't, and which files aren't being tracked by Git. Status output does not show any information regarding the committed project history. What is the result of issuing this command?

```
villareal1@workstation:~/CPE232_villareal$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
```

j. Use the command git add README.md to add the file into the staging area.

```
villareal1@workstation:~/CPE232_villareal$ git add README.md
```

k. Use the git commit -m "your message" to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next commit.

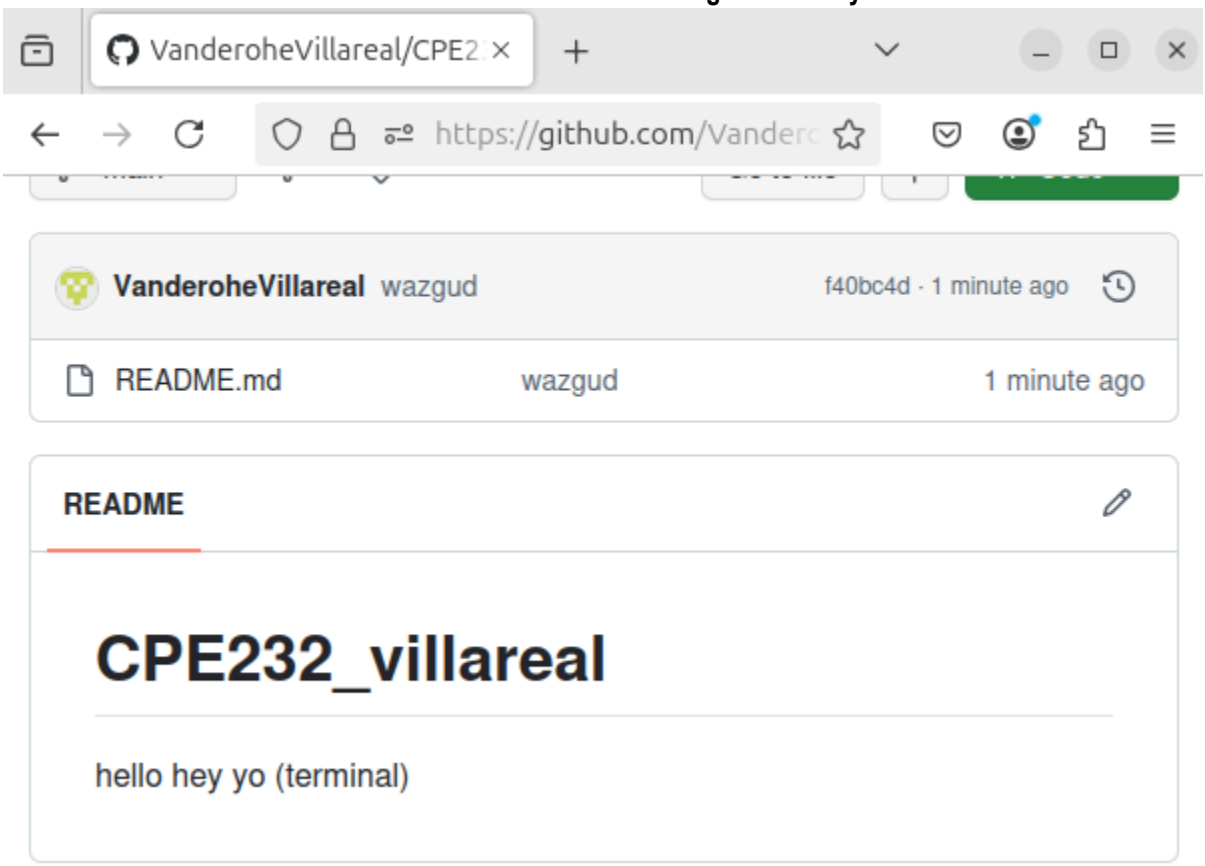
```
villareal1@workstation:~/CPE232_villareal$ git commit -m "wazgud"
[main f40bc4d] wazgud
 1 file changed, 3 insertions(+), 1 deletion(-)
```

l. Use the command git push <remote><branch> to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository.

As an example, you may issue git push origin main.

```
villareal1@workstation:~/CPE232_villareal$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 286 bytes | 286.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:VanderoheVillareal/CPE232_villareal.git
   a80e2dc..f40bc4d  main -> main
```

m. On the GitHub repository, verify that the changes have been made to README.md by refreshing the page. Describe the README.md file. You can notice how long the last commit was. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

1. How will you describe the ssh-program? What does it do?

- The ssh-program is for connecting a local machine or workstation to a remote server where the user can configure that remote server using the local machine.

2. How do you know that you already installed the public key to the remote servers?

- By using the cat ~/.ssh/authorized_keys command, this way if the command prints a text containing random text characters this verifies that the public key is existing on the remote server. If not it would only

print no file or directory exists.

3. What sort of things have we so far done to the remote servers using ansible commands?
- We changed the contents of the README.md using the terminal of my workstation. I used the existing git repository and cloned it to be able to configure using the terminal of the ubuntu and by using the same public key.

4. How important is the inventory file?
- This is important because in infrastructure automation inventory file is the one that defines the target servers, groups, and the connection parameters.

Conclusions/Learnings:

In conclusion of this activity I learned that there are ways to configure a github repository using the ubuntu's terminal where connecting it wasn't as confusing as the first two tasks. The first two tasks made me create an SSH key pair (using ssh-keygen) for the user authentication that would eventually be used for configuring the remote server using the local machine or the workstation. We then copied the key to the remote server in order to allow us to configure it from the workstation. In other words, this activity made me understand a lot of problems encountered like being unable to run the ssh command even though the command for copying the public key to the remote servers is already done, this made me redo the command once again with a confirmation this time, another is learn from past mistakes and avoid the same problems encountered.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Idea generation, solving errors**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Ai is used to assist me in constructing an answer and help find ways to fix encountered errors/problems.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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